

MILESTONE 4

Backend Integration & End-to-End System Integration

Enterprise Learning Platform – Skill & Career Guidance System

1. Milestone Overview

Milestone 4 focuses on the **complete integration of the frontend application with the backend services and database**.

The interfaces and dashboards developed in Milestones 1–3 are connected to the actual backend APIs, authentication services, database, and business logic.

This milestone transforms the previously developed UI screens into a **fully connected and functional application**.

The main objective is to ensure that data entered or generated through the frontend is processed by the backend, stored in the database, and retrieved dynamically whenever required.

2. Milestone 4 Objectives

The integration includes:

- Frontend-to-backend API integration
- Database integration
- JWT authentication integration
- Role-based authorization integration
- User management integration
- Student dashboard integration
- Admin dashboard integration
- Mentor dashboard integration
- Course management integration
- Enrollment integration
- Assignment integration

- Quiz integration
 - Session integration
 - Achievement integration
 - Certificate integration
 - Notification integration
 - Complaint integration
 - Approval integration
 - Audit log integration
 - Search integration
 - Error handling
 - Loading states
 - End-to-end data flow
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3. Frontend–Backend Integration

All major frontend screens are connected to their corresponding backend APIs.

Instead of using static or mock data, the application retrieves and updates information through real backend services.

Integration Flow

Frontend UI

↓

API Request

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Backend Controller / Route

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Business Logic / Service Layer

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Database

↓

Backend Response



Frontend State Update



Updated UI

This architecture ensures that the information displayed to users represents the actual application data.

4. Database Integration

The backend is connected to the application's database.

Database integration supports persistent storage for:

- Users
- Roles
- Courses
- Lessons
- Enrollments
- Assignments
- Quizzes
- Quiz results
- Learning progress
- Achievements
- Certificates
- Notifications
- Complaints
- Internships
- Sessions
- Audit logs
- Other platform data

Data created or modified through the application is persisted in the database.

5. Authentication Integration

The JWT authentication system developed in Milestone 1 is fully integrated with the backend.

Authentication Flow

Login / Registration



Backend Authentication API



Database Verification



JWT Generation



Frontend Authentication State



Protected API Requests



JWT Verification



Authenticated User Data

This ensures that users can securely access their respective platform areas.

6. Role-Based Authorization Integration

The backend determines the authenticated user's role and applies appropriate authorization rules.

Supported roles include:

- Student

- Mentor
- Admin

Access Control

Student

→ Student Dashboard and student services

Mentor

→ Mentor Dashboard and teaching services

Admin

→ Admin Dashboard and platform management services

Unauthorized users are prevented from accessing restricted APIs and resources.

7. Student Dashboard Integration

The Student Dashboard developed in Milestone 2 is connected to live backend data.

The dashboard retrieves:

- Student profile
- Learner XP
- Active courses
- Completed courses
- Study hours
- Learning streak
- Achievements
- Certificates
- Course progress
- Weekly progress
- Notifications
- Recommendations

The values displayed on the dashboard are dynamically generated from the backend and database.

8. Admin Dashboard Integration

The Admin Dashboard developed in Milestone 3 is connected to live backend services.

The dashboard retrieves:

- Total users
- Students
- Mentors
- Administrators
- Total courses
- Active courses
- Pending approvals
- Complaints
- Audit logs
- Notifications
- Platform activity

Administrative actions performed through the dashboard are sent to the backend and persisted where applicable.

9. Mentor Dashboard Integration

The Mentor Dashboard is connected to backend services for real-time teaching and course-management operations.

The dashboard retrieves:

- Total students
- Courses created
- Published courses
- Draft courses
- Pending assignments
- Pending quizzes
- Total enrollments
- Recent student activity
- Notifications
- Upcoming sessions

Mentor actions such as creating courses, managing assignments, and scheduling sessions are processed by the backend.

10. User Management Integration

The user management system is integrated with the backend database.

Supported operations include:

- Create user
- View users
- Search users
- Update user information
- Manage user status
- Assign roles
- View user details

Changes made by administrators are reflected across the platform.

11. Course Management Integration

Course management is connected to backend APIs.

The system supports:

- Course creation
- Course editing
- Course deletion where permitted
- Course drafts
- Course submission
- Course approval
- Course rejection
- Course publishing
- Course enrollment
- Course progress

Course Lifecycle

Create



Draft



Submit



Pending Approval



Admin Review



Approved / Rejected



Published

12. Student Enrollment Integration

Students can enroll in available courses through the frontend.

The enrollment process is:

Student selects course



Enrollment API



Backend validates request



Enrollment stored in database



Student dashboard updated



Course becomes available to student

Mentors and administrators can view relevant enrollment information.

13. Learning Progress Integration

Course progress is dynamically connected to the backend.

The system can track:

- Course completion
- Lesson completion
- Learning activity
- Progress percentage
- Study hours
- Course status

When a student completes learning activities, the backend updates the corresponding progress data.

The Student Dashboard then reflects the updated information.

14. Assignment Integration

Assignment functionality is connected across Student and Mentor interfaces.

Student

Students can:

- View assignments
- Submit assignments
- Track submission status

Mentor

Mentors can:

- Create assignments
- View submissions
- Review submissions
- Grade assignments
- Provide feedback

Assignment data is stored and retrieved through backend APIs.

15. Quiz Integration

Quiz functionality is integrated with the backend.

The system supports:

- Quiz creation
- Quiz management
- Quiz questions
- Quiz attempts
- Quiz submission
- Result calculation
- Score storage
- Mentor review
- Student result display

Quiz activity can also contribute to student progress and achievements.

16. Session Integration

Mentors can create and manage learning or mentorship sessions through backend APIs.

The system supports:

- Create session
- Update session
- Schedule session
- View upcoming sessions
- Session information

- Student access to relevant sessions

Session information is stored persistently in the database.

17. Achievement Integration

Achievements are connected to student activity.

The backend can determine when a student meets achievement conditions.

Examples include:

- First Enrollment
- Course Finisher
- Learning Streak
- Quiz Achievement
- Skill Achievement

When an achievement is earned, the backend records it and the Student Dashboard displays it.

18. Certificate Integration

Certificate-related functionality is integrated with student course completion.

The system can:

- Determine course completion
- Generate/store certificate information
- Associate certificates with students
- Display certificates
- Track certificate status

Certificates can subsequently support QR verification and credential validation.

19. Notification Integration

The notification system is connected to backend events.

Notifications can be generated for:

- New course enrollment
- Assignment submission
- Quiz activity
- Course approval
- Course rejection
- New session
- Achievement earned
- Certificate availability
- Platform announcements
- Administrative events

Unread/read status is maintained through the backend.

20. Complaint Integration

The Admin Complaint Desk is connected to backend complaint services.

The system supports:

- Complaint creation
- Complaint listing
- Complaint details
- Status updates
- Resolution
- Complaint history

Supported statuses may include:

- Open
- In Review
- Resolved
- Closed

Administrative updates are persisted in the database.

21. Approval Integration

Course approval functionality is fully connected to backend services.

Process

Mentor submits course

↓

Backend stores pending status

↓

Admin receives notification

↓

Admin reviews course

↓

Admin approves/rejects

↓

Backend updates course status

↓

Mentor receives updated status

This creates a complete mentor-to-admin course publishing workflow.

22. Audit Log Integration

Administrative and important system activities are recorded through the audit system.

Events may include:

- User changes
- Role changes
- Course approvals

- Course rejections
- Complaint updates
- Administrative actions
- System events

Audit information can include:

- User
- Action
- Resource
- Timestamp
- Description

This provides traceability across the platform.

23. Search Integration

The global search interface is connected to backend search functionality.

Users can search relevant platform content such as:

- Courses
- Roadmaps
- Mentors
- Learning resources

Search results are retrieved dynamically based on the user's query.

24. API Integration

The frontend communicates with backend APIs for all major operations.

API integration covers:

- Authentication APIs
- User APIs
- Course APIs
- Enrollment APIs
- Progress APIs

- Assignment APIs
- Quiz APIs
- Achievement APIs
- Certificate APIs
- Notification APIs
- Session APIs
- Complaint APIs
- Approval APIs
- Audit APIs
- Search APIs
- Dashboard APIs

All protected requests use the authentication mechanism established in previous milestones.

25. Error Handling

The application includes frontend and backend error handling.

The system should handle:

- Invalid requests
- Authentication failures
- Authorization failures
- Validation errors
- Missing resources
- Database errors
- Network/API errors
- Expired authentication
- Duplicate records
- Server errors

Users should receive clear and meaningful messages without exposing sensitive backend information.

26. Loading & Empty States

Frontend components should provide appropriate states while communicating with backend services.

Loading State

Displayed while data is being retrieved.

Empty State

Displayed when no data is available.

Examples:

- No courses enrolled
- No notifications
- No assignments
- No achievements
- No upcoming sessions

Error State

Displayed when data cannot be retrieved successfully.

27. Data Consistency

The system should maintain consistency between frontend and backend data.

For example:

When a student enrolls in a course:

Enrollment Created

↓

Database Updated

↓

Student Dashboard Updated

↓

Active Course Count Updated

↓

Course Appears in My Enrolled Courses

Similarly, when a mentor publishes a course:

Course Submitted

↓

Admin Approval

↓

Course Status Updated

↓

Course Becomes Available

↓

Students Can Enroll

28. Security Integration

Milestone 4 integrates all previously established security mechanisms.

The system includes:

- JWT authentication
 - Password hashing
 - Role-based authorization
 - Protected API routes
 - Input validation
 - Secure API communication
 - Authentication expiry
 - Permission checking
 - Protected user data
 - Secure database access
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29. End-to-End Integration

The platform now supports complete end-to-end workflows.

Student Workflow

Register



Login



JWT Authentication



Student Dashboard



Browse Courses



Enroll



Learn



Complete Lessons



Complete Quiz / Assignment



Update Progress



Earn Achievement



Receive Certificate

Mentor Workflow

Login



JWT Authentication



Mentor Dashboard



Create Course



Save Draft



Submit for Approval



Admin Review



Course Published



Students Enroll



Monitor Student Activity



Review Assignments / Quizzes



Manage Sessions

Admin Workflow

Login



JWT Authentication



Admin Dashboard



Monitor Platform



Manage Users



Review Course Approvals



Manage Complaints



Monitor Audit Logs



Manage Platform Operations

30. Milestone 4 Deliverables

Frontend Integration

- All dashboard APIs connected
- Authentication connected
- User APIs connected
- Course APIs connected
- Enrollment APIs connected
- Progress APIs connected
- Assignment APIs connected
- Quiz APIs connected
- Achievement APIs connected
- Certificate APIs connected
- Notification APIs connected
- Session APIs connected
- Complaint APIs connected
- Approval APIs connected
- Audit APIs connected
- Search APIs connected

Backend Integration

- API routes implemented
- Controllers/services connected
- Database connected
- Authentication middleware connected
- Authorization middleware connected
- Business logic integrated
- Validation integrated
- Error handling integrated

Database

- User data persistence
- Course data persistence
- Enrollment persistence
- Learning progress persistence
- Assignment persistence
- Quiz persistence
- Achievement persistence

- Certificate persistence
 - Notification persistence
 - Session persistence
 - Complaint persistence
 - Audit log persistence
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31. Milestone 4 Outcome

At the completion of Milestone 4, the platform moves from primarily interface-level functionality to a **fully integrated application**.

The frontend, backend, database, authentication, authorization, and business logic operate together as one system.

The platform supports complete workflows for:

Students + Mentors + Administrators

with real-time data flowing between:

UI → API → Backend → Database → API → UI

Final Milestone Statement

Milestone 4 completes the backend integration of the Enterprise Learning Platform, connecting all major frontend modules, dashboards, authentication, authorization, APIs, business logic, and database services into a functional end-to-end system.